Math 675 Spring 2026:
Homework 4 (LATE)
Due Tuesday, March 3rd, 2026

Question 1:

a) $\mathbf{x}' = \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix} \mathbf{x}$

Finding the general Solution

$$\begin{vmatrix} 3-\lambda & -4 \\ 1 & -1-\lambda \end{vmatrix} = (3-\lambda)(-1-\lambda) - (-4)(1) = 0$$

$$= -3 - 3\lambda + \lambda + \lambda^2 + 4 = 0$$

$$= \lambda^2 - 2\lambda + 1 = 0$$

$$= (\lambda - 1)^2 = 0$$

$$\Rightarrow \lambda_1 = \lambda_2 = 1$$

Repeated Eigenvalue #1 and #2

$\lambda_{\mathbf{1}} = \quad \mathbf{1}$: Solve for $(A - \lambda I)\mathbf{v} = 0$

$$\begin{bmatrix} 3-\lambda_1 & -4 \\ 1 & -1-\lambda_1 \end{bmatrix} \to \begin{bmatrix} 2 & -4 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} 2v_1 - 4v_2 &= 0 \\ 1v_1 - 2v_2 &= 0 \end{aligned} \longrightarrow \quad v_1 = 2v_2 \quad \longrightarrow \mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Solve for $(A - \lambda I)\mathbf{w} = \mathbf{v}$

$$\begin{bmatrix} 3-\lambda_1 & -4 \\ 1 & -1-\lambda_1 \end{bmatrix} \to \begin{bmatrix} 2 & -4 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\begin{aligned} 2w_1 - 4w_2 &= 2 \\ 1w_1 - 2w_2 &= 1 \end{aligned} \longrightarrow \quad w_1 = 1 + 2w_2 \quad \longrightarrow \mathbf{w} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

So our general solution is $\mathbf{x}(t) = c_1 e^t \begin{pmatrix} 2 \\ 1 \end{pmatrix} + c_2 e^t \left[t \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right]$

Describing the behavior of the system

As $t \to \infty$, we can see that e^t approaches infinity in both of the terms. But the te^t term grows faster than e^t term, so solutions will become parrellel to the lint $y = \frac{1}{2}x$ as $t \to \infty$.

$$\lim_{t\to\infty} \frac{c_1 e^t v}{c_2 e^t (tv + w)} = \lim_{t\to\infty} \frac{c_1 v}{c_2 (tv + w)} = 0$$

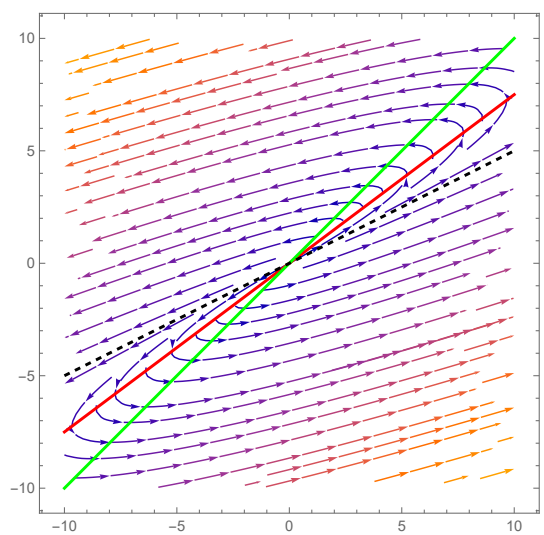
Sketching the phase portrait

We want to find the nullclines of the system, so we set each derivative to 0:

Set $x' = 0$: so $0 = 3x - 4y$, giving us $y = \frac{3}{4}x$
Set $y' = 0$: so $0 = x - y$, giving us $y = x$

We still also have our initial eigenvalue direction of $y = \frac{1}{2}x$

Using technology to draw the phase portrait



- The black dashed line is the eigendirection $y = \frac{1}{2}x$, which is the direction that solutions will become parallel to as $t \to \infty$
- The red line is the x-nullcline $y = \frac{3}{4}x$. Trajectories cross this line vertically and above this line, $x' < 0$, and below it, $x' > 0$.
- The green line is the y-nullcline $y = x$. Trajectories cross this line horizontally and above this line, $y' < 0$, and below it, $y' > 0$.

b) $\mathbf{x}' = \begin{pmatrix} 4 & -2 \\ 8 & -4 \end{pmatrix} \mathbf{x}$

Finding the general Solution

$$\begin{vmatrix} 4-\lambda & -2 \\ 8 & -4-\lambda \end{vmatrix} = (4-\lambda)(-4-\lambda) - (-2)(8) = 0$$

$$= -16 - 4\lambda + 4\lambda + \lambda^2 + 16 = 0$$

$$= \lambda^2 = 0$$

$$\Rightarrow \lambda_1 = \lambda_2 = 0$$

Repeated Eigenvalue #1 and #2

$\lambda_{\mathbf{1}} = \quad \mathbf{0}$: Solve for $(A - \lambda I)\mathbf{v} = 0$

$$\begin{bmatrix} 4-\lambda_1 & -2 \\ 8 & -4-\lambda_1 \end{bmatrix} \to \begin{bmatrix} 4 & -2 \\ 8 & -4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} 4v_1 - 2v_2 &= 0 \\ 8v_1 - 4v_2 &= 0 \end{aligned} \longrightarrow \quad v_2 = 2v_1 \quad \longrightarrow \mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Solve for $(A - \lambda I)\mathbf{w} = \mathbf{v}$

$$\begin{bmatrix} 4-\lambda_1 & -2 \\ 8 & -4-\lambda_1 \end{bmatrix} \to \begin{bmatrix} 4 & -2 \\ 8 & -4 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\begin{aligned} 4w_1 - 2w_2 &= 1 \\ 8w_1 - 4w_2 &= 2 \end{aligned} \longrightarrow \quad w_1 = \frac{1}{4} + \frac{1}{2}w_2 \quad \longrightarrow \mathbf{w} = \begin{bmatrix} \frac{1}{4} \\ 0 \end{bmatrix}$$

So our general solution is $\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + c_2 \left[t \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} \frac{1}{4} \\ 0 \end{pmatrix} \right]$

Describing the behavior of the system

As $t \to \infty$, we can see that the single $c_2 t$ term approaches infinity.
But the first term does not change at all, so solutions will become parrellel to the line $y = 2x$ as $t \to \infty$.

$$\lim_{t\to\infty} \frac{c_1 v}{c_2(tv + w)} = 0$$

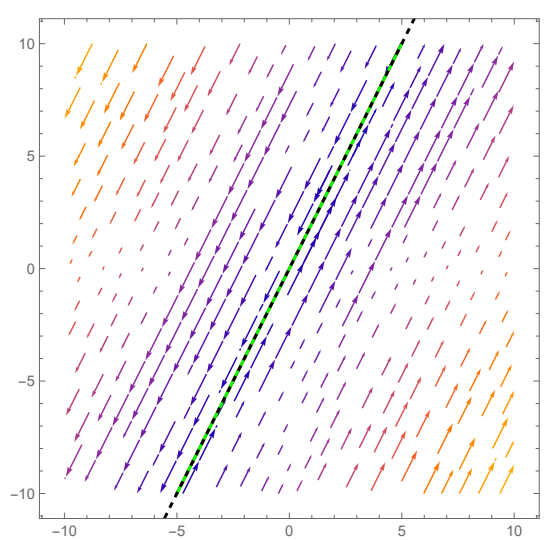
Sketching the phase portrait

We want to find the nullclines of the system, so we set each derivative to 0:

Set $x' = 0$: so $0 = 4x - 2y$, giving us $y = 2x$
Set $y' = 0$: so $0 = 8x - 4y$, giving us $y = 2x$

We still also have our initial eigenvalue direction of $y = 2x$ This means that instead of two nullclines, we only have one nullcline.

Using technology to draw the phase portrait



- The black dashed line is the eigendirection $y = 2x$,
  which is the direction that solutions will become parallel to as $t \to \infty$
- The green line is the y-nullcline $y = 2x$. Trajectories do not cross this line. bove this line, $y' < 0$ and $x' < 0$, and below it, $y' > 0$ and $x' > 0$.

c) $\mathbf{x}' = \begin{pmatrix} -\frac{3}{2} & 1 \\ -\frac{1}{4} & -\frac{1}{2} \end{pmatrix} \mathbf{x}$

Finding the general Solution

$$\begin{vmatrix} -\frac{3}{2} - \lambda & 1 \\ -\frac{1}{4} & -\frac{1}{2} - \lambda \end{vmatrix} \rightarrow trace(A) = -\tfrac{3}{2} + (-\tfrac{1}{2}) = -2$$

$$\rightarrow det(A) = (-\tfrac{3}{2})(-\tfrac{1}{2}) - (1)(-\tfrac{1}{4}) = \tfrac{3}{4} + \tfrac{1}{4} = 1$$

$$\rightarrow \lambda^2 - trace(A)\lambda + det(A) = \lambda^2 + 2\lambda + 1 = 0$$

$$\rightarrow (\lambda + 1)^2 = 0$$

$$\Rightarrow \lambda_1 = \lambda_2 = -1$$

Repeated Eigenvalue #1 and #2

$\lambda_{\mathbf{1}} = -\mathbf{1}$: Solve for $(A - \lambda I)\mathbf{v} = 0$

$$\begin{bmatrix} -\frac{3}{2} - \lambda_1 & 1 \\ -\frac{1}{4} & -\frac{1}{2} - \lambda_1 \end{bmatrix} \rightarrow \begin{bmatrix} -\frac{1}{2} & 1 \\ -\frac{1}{4} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} -\tfrac{1}{2}v_1 + 1v_2 &= 0 \\ -\tfrac{1}{4}v_1 + \tfrac{1}{2}v_2 &= 0 \end{aligned} \longrightarrow \quad v_1 = 2v_2 \quad \longrightarrow \mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Solve for $(A - \lambda I)\mathbf{w} = \mathbf{v}$

$$\begin{bmatrix} -\frac{3}{2} - \lambda_1 & 1 \\ -\frac{1}{4} & -\frac{1}{2} - \lambda_1 \end{bmatrix} \rightarrow \begin{bmatrix} -\frac{1}{2} & 1 \\ -\frac{1}{4} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\begin{aligned} -\tfrac{1}{2}w_1 + 1w_2 &= 2 \\ -\tfrac{1}{4}w_1 + \tfrac{1}{2}w_2 &= 1 \end{aligned} \longrightarrow \quad w_1 = 2w_2 - 4 \quad \longrightarrow \mathbf{w} = \begin{bmatrix} -4 \\ 0 \end{bmatrix}$$

So our general solution is $\mathbf{x}(t) = c_1 e^{-t} \begin{pmatrix} 2 \\ 1 \end{pmatrix} + c_2 e^{-t} \left[t \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} -4 \\ 0 \end{pmatrix} \right]$

Describing the behavior of the system

As $t \to \infty$, we can see that e^{-t} approaches 0 in both of the terms. But the e^{-t} term approaches 0 faster than te^{-t} term, so solutions will become parallel to the line $y = \frac{1}{2}x$ as $t \to \infty$.

$$\lim_{t\to\infty} \frac{c_1 e^{-t} v}{c_2 e^{-t}(tv + w)} = \lim_{t\to\infty} \frac{c_1 v}{c_2(vt + w)} = 0$$

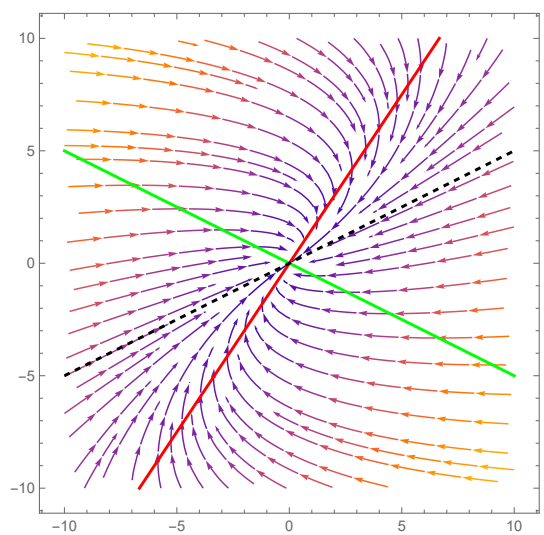
Sketching the phase portrait

We want to find the nullclines of the system, so we set each derivative to 0:

Set $x' = 0$: so $0 = -\frac{3}{2}x + 1y$, giving us $y = \frac{3}{2}x$
Set $y' = 0$: so $0 = -\frac{1}{4}x - \frac{1}{2}y$, giving us $y = -\frac{1}{2}x$

We still also have our initial eigenvalue direction of $y = \frac{1}{2}x$

Using technology to draw the phase portrait



- The black dashed line is the eigendirection $y = \frac{1}{2}x$, which is the direction that solutions will become parallel to as $t \to \infty$
- The red line is the x-nullcline $y = \frac{3}{2}x$. Trajectories cross this line vertically, and above this line, $x' > 0$, and below it, $x' < 0$.
- The green line is the y-nullcline $y = -\frac{1}{2}x$. Trajectories cross this line horizontally, and above this line, $y' < 0$, and below it, $y' > 0$.

Question 2:

Find the general solution of the 3D System:

$$\mathbf{x}'(t) = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 0 & -1 & 1 \end{pmatrix} \mathbf{x}(t)$$

$$\begin{aligned}
\begin{vmatrix} 1-\lambda & 1 & 1 \\ 2 & 1-\lambda & -1 \\ 0 & -1 & 1-\lambda \end{vmatrix} &= (1-\lambda)\begin{vmatrix} 1-\lambda & -1 \\ -1 & 1-\lambda \end{vmatrix} - 1\begin{vmatrix} 2 & -1 \\ 0 & 1-\lambda \end{vmatrix} + 1\begin{vmatrix} 2 & 1-\lambda \\ 0 & -1 \end{vmatrix} \\
&= (1-\lambda)\left[(1-\lambda)^2 - 1\right] - 1[2(1-\lambda)] + 1[-2] = 0 \\
&= (1-\lambda)(-2\lambda + \lambda^2) + (-2 + 2\lambda) + (-2) = 0 \\
&= -2\lambda + \lambda^2 + 2\lambda^2 - \lambda^3 - 2 + 2\lambda - 2 \\
&= -\lambda^3 + 3\lambda^2 - 4 = 0 \\
&= \lambda^3 - 3\lambda^2 + 4 = 0
\end{aligned}$$

Now this is where the tricky bit is, factoring a cubic can be long and tedious so lets use polynomial long division, or better yet, synthetic division. Notice that plugging in $\lambda = -1$ gives us 0, so we know that $\lambda + 1$ is a factor of our cubic.

$$\begin{array}{r|rrrr}
-1 & 1 & -3 & 0 & 4 \\
 & & 1 & 4 & -4 \\
\hline
 & 1 & -4 & 4 & \boxed{0}
\end{array}$$

This now tells us that our cubic can be factored into $-(\lambda+1)(\lambda^2 - 4\lambda + 4)$
Now we have to factor a quadratic, which is super easy

$$-(\lambda+1)(\lambda^2 - 4\lambda + 4) = (\lambda+1)(\lambda-2)^2 = 0$$

Thus, the eigenvalues for A are: $\lambda_1 = -1, \lambda_2 = \lambda_3 = 2$

Eigenvalue #1

$\lambda_{\mathbf{1}} = -\mathbf{1}$: Solve for $(A - \lambda I)\mathbf{v} = 0$

$$\begin{bmatrix} 1-\lambda_1 & 1 & 1 \\ 2 & 1-\lambda_1 & -1 \\ 0 & -1 & 1-\lambda_1 \end{bmatrix} = \left[\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 2 & 2 & -1 & 0 \\ 0 & -1 & 2 & 0 \end{array}\right] \xrightarrow{\text{RREF}} \left[\begin{array}{ccc|c} 1 & 0 & \frac{3}{2} & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\text{This gives us that } \begin{cases} a &= \frac{3}{2}c \\ b &= -c \\ c &= c \end{cases} \text{ so our eigenvalue is } \mathbf{v}_1 = \begin{bmatrix} 3 \\ -4 \\ 2 \end{bmatrix}$$

Repeated Eigenvalue #2 and #3

$\lambda_{\mathbf{2}} = \quad \mathbf{2}$: Solve for $(A - \lambda I)\mathbf{v} = 0$

$$\begin{bmatrix} 1-\lambda_2 & 1 & 2 \\ 1 & 1-\lambda_2 & 1 \\ 2 & 1 & 1-\lambda_2 \end{bmatrix} = \left[\begin{array}{ccc|c} -1 & 1 & 1 & 0 \\ 2 & -1 & -1 & 0 \\ 0 & -1 & -1 & 0 \end{array}\right] \xrightarrow{\text{RREF}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\text{This gives us that } \begin{cases} a &= 0 \\ b &= -c \\ c &= c \end{cases} \text{ so our eigenvalue is } \mathbf{v}_2 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

Solve for $(A - \lambda I)\mathbf{w} = \mathbf{v}$

$$\begin{bmatrix} 1-\lambda_2 & 1 & 2 \\ 1 & 1-\lambda_2 & 1 \\ 2 & 1 & 1-\lambda_2 \end{bmatrix} = \left[\begin{array}{ccc|c} -1 & 1 & 1 & 0 \\ 2 & -1 & -1 & -1 \\ 0 & -1 & -1 & 1 \end{array}\right] \xrightarrow{\text{RREF}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\text{This gives us } \begin{cases} w_1 &= -1 \\ w_2 &= -1 \\ w_3 &= 0 \end{cases} \text{ so our generalized eigenvalue is } \mathbf{w} = \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix}$$

So our general solution is $\mathbf{x}(t) = c_1 e^{-t} \begin{pmatrix} 3 \\ -4 \\ 2 \end{pmatrix} + c_2 e^{2t} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} + c_3 e^{2t} \left[t \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix} \right]$